

CS480P/580P Introduction to Natural Language Processing

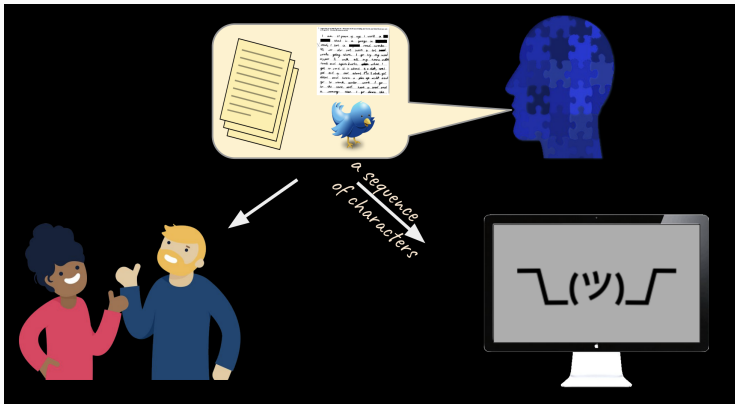
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August 19, 2025

Lecture 1

Natural language is complicated!

Natural language is complicated!



Natural language is complicated!

What is natural language like for a computer?

The horse raced past the barn.



Natural language is complicated!

What is natural language like for a computer?

The horse raced past the barn.

The horse raced past the barn fell.



Natural language is complicated!

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The horse raced past the barn.



The horse raced past the barn fell.



Natural language is complicated!

What is natural language like for a computer?

The horse raced past the barn. ✓

The horse raced past the barn fell. ✓

The horse **runs** past the barn. ✓

The horse **runs** past the barn fell. ✗

Natural language is complicated!

More empathy for the computer...

She ate the cake with the frosting.



Natural language is complicated!

More empathy for the computer...

She ate the cake with the frosting.

`['She', 'ate', X, 'with', Y, '.']`



Natural language is complicated!

More empathy for the computer...

She ate the cake with the frosting.



```
['She', 'ate', X, 'with', Y, '.']  
=> Y is a part of X
```

Natural language is complicated!

More empathy for the computer...

She ate the cake with the frosting.

She ate the cake with the fork.

['She', 'ate', X, 'with', Y, '.']

=> Y is a ~~part~~ of X



Natural language is complicated!

More empathy for the computer...

She ate the cake with the frosting.

She ate the cake with the fork.



Natural language is complicated!

More empathy for the computer...

She ate the cake with the frosting.

She ate the cake with the fork.

He put the **port** on the ship.

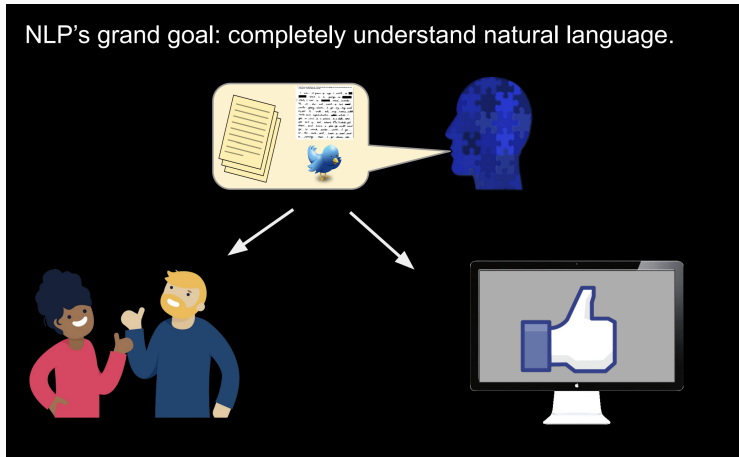
He walked along the **port** of the ship.

He walked along the **port** next to the ship.



Natural language is complicated!

NLP's grand goal: completely understand natural language.



NLP's practical applications



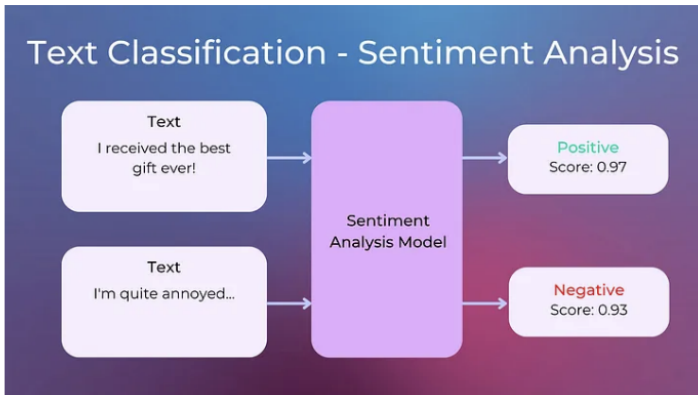
- Machine translation
- Sentiment Analysis
- Automatic speech recognition
 - Personalized assistants
 - Auto customer service
- Information Retrieval
 - Web Search
 - Question Answering
- Computational Social Science
- *Growing day by day*

NLP Topics



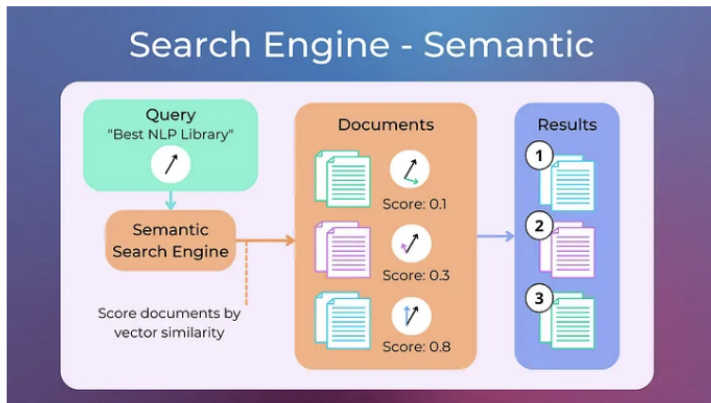
<https://medium.com/nlplanet/what-tasks-can-i-solve-with-nlp-today-1b1823cc8cdf>

- **Sentiment Analysis:** Understanding whether a text has a positive sentiment (e.g. “the dinner was nice”) or a negative sentiment (e.g. “the dinner was awful”).

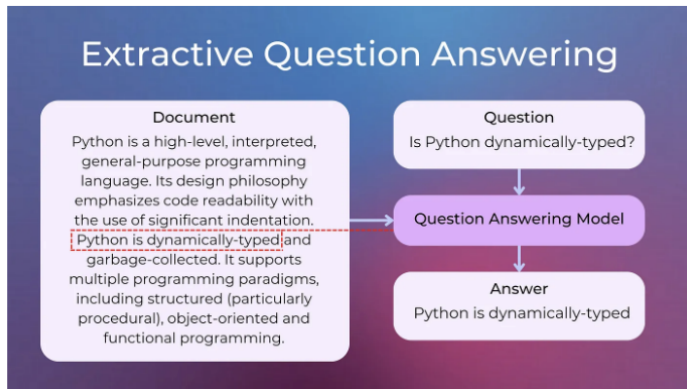


- **Spam Detection:** Moving the right emails into the spam folders.
- **Assigning categories to CRM customer support tickets:** Detecting the type of a ticket (e.g. refund or technical problem) and dispatching it automatically to the correct customer support person.
- **Classification of disaster conversations on social media:** Detecting disasters like fires or floods from social media posts.

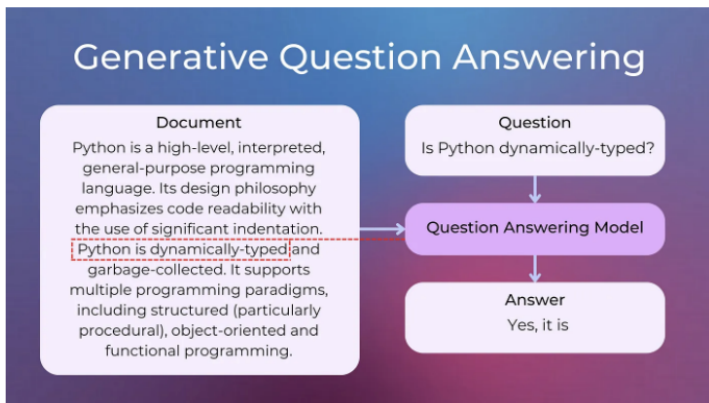
Semantic search is related to information retrieval in that it is concerned with finding the best match for a user's query, but it goes beyond traditional IR techniques by considering the semantic meaning of the queries and the documents thanks to word embeddings.



- **Extractive Question Answering** takes a question as input and retrieves the most relevant answer from a large database of potential answers.



- **Generative Question Answering**, on the other hand, generates an answer from scratch based on the question and sometimes also on additional context information.



Solved by Machine Learning

NLP's practical applications



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how?
→

- Machine learning:
 - Logistic regression
 - Probabilistic modeling
 - Recurrent Neural Networks
 - Transformers
- Algorithms, e.g.:
 - Graph analytics
 - Dynamic programming
- Data science
 - Hypothesis testing